

TEACHING AND EXAMINATION SCHEME:

Teaching Scheme					Examination Scheme								
Hrs / week				Credits	TH Paper Hrs.	Marks							
TH	TU	PR	HRS				TH	TEST	TH+TEST	PR	OR	TW	TOTAL
03	--	02	05	04	03	Max.	80	20	100	25	--	25	150
						Min.	32	--	40	10	--	10	--

1.0 RATIONALE:

In the current era of networking, online transaction processing and managing the dataflow over network becomes an important issue. This course is essential for providing knowledge and hands on experience over the issues of managing data on Web, developing powerful GUI based friendly user interface, server side programming and developing applications for communication over network using object oriented fundamentals. Advanced Java enhances the Java programming. After learning this course, the student will be able to develop standalone/network/web based software projects required in curriculum as well as industry

2.0 COURSE OBJECTIVES:

The student will be able to,

1. Develop applets, Frames using awt and swing components.
2. Develop network based applications.
3. Perform CRUD operation on database.
4. Develop server side programs.
5. Develop small desktop or web applications.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Design and Develop applets, Frames using awt and swing components.
2. Develop network based applications.
3. Use CURD operation in standalone and web applications.
4. Use server side programs for developing web applications.
5. Develop servlet program

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I Introduction to Abstract Window Toolkit (AWT) & Swings	1a. Frame and Applet 1b. Describe AWT controls and Layout Managers 1c. Introduction to Swing	1.1 Component, container, window, frame, panel, Creating windowed programs & applets. 1.2 AWT controls & layout managers 1.3 Understanding the use of AWT controls: labels, buttons, checkbox, checkbox group, scroll bars, text field, text area 1.4 Understanding the use of layout managers: flow Layout, BorderLayout, GridLayout, cardLayout, gridbagLayout, menubars, menus, dialog boxes, file dialog. 1.5 Introduction to swing, Swing features,	10

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		MVC Architecture, Combo Boxes, progress bar,	
Unit-II Event Handling	2a. Introduction to Delegation Event Model 2b. Explain Event Listener Interfaces.	2.1 The delegation Event Model, Event sources, Event listeners, Event classes. The Action Event class, The Component Event class, the Container Event class, the Focus Event class, the Item Event class, the Key Event class, the Mouse Event class, the Text Event class, the Window Event class, Adapter classes, Inner classes 2.2 Event listener interfaces, The ActionListener Interface, the ComponentListener Interface, the ContainerListener Interface, the FocusListener Interface, the ItemListener Interface, the KeyListener Interface, the MouseListener Interface, the MouseMotion Interface, the TextListener, Interface, the WindowListener Interface, the WindowFocusListener Interface	06
Unit-III Networking & Security	3a. Basics of Networking 3b. Explain InetAddress Class 3c. Describe Sockets 3d. Explain URL class 3e. Security	3.1 Basics of Networking: Socket, IP, TCP, UDP, Proxy Server, Internet Addressing 3.2 The InetAddress Class Factory methods, Instance methods 3.3 TCP/IP Sockets, Socket, Server Socket, methods 3.4 URL, URL Connection, http, URL Connection methods, creating & using TCP/IP client & server 3.5 Security with Java: Theoretical introduction to java security, secure coding guidelines for java programming language.	08
Unit-IV Interacting with Database	4e. Explain Connecting JDBC-ODBC	4.1 JDBC, ODBC, & Other APIS, JDBC two tier & three tier models, connecting to Database. Driver Interface, Driver Manager Class, Connection Interface, Statement Interface, the java.sql.package. Establishing connection & retrieving Information, ResultSetInterface, CRUD operations using JDBC	10
Unit-V Servlets, JSP & Spring	5a. Introduction to Servlet 5b. Describe Session and Cookies 5c. Introduction to JSP	5.1 Basics of Web applications, use of tomcat server, Type of Servlet, Servlet life cycle, using servlets, handling request and response. 5.2 Basic concepts of sessions, cookies & session tracking.	14

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		5.3 JSP introduction, life cycle of JSP, scriptlet tag, expression tag, declaration tag, Implicit Objects, Directives, Action Elements, JSP Expression language, JSTL, JSP custom tags.	
		TOTAL	48

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	Introduction to Abstract Windowing Toolkit(AWT) & Swings	06	06	04	16
II	Event Handling	02	04	04	10
III	Networking & Security	06	04	04	14
IV	Interacting with Database	04	04	08	16
V	Servlets, JSP & Spring	04	08	12	24
	TOTAL	22	26	32	80

Legends: R = Remembrance (Knowledge); U= Understanding; A= Application and above levels (Revised Bloom's taxonomy)

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

6.0 ASSIGNMENTS/TUTORIALS/PRACTICALS/TASKS:

The tutorial/practical/assignments/tasks should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the competencies.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in affective domain** as given in a common list at the beginning of curriculum document for this programme. Faculty should refer to that common list and should ensure that students also acquire those Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	Practical Exercises (Outcomes in Psychomotor Domain)	Approx. Hrs. Required
01	I	Write a program using AWT to design a form using the components text field, label, checkbox, button, and list.	04*
02	II	Write a program to design a calculator using Java components and handle various events related to each component and apply proper layout to it.	04
03	I, II	Write a program to create a menu bar with various menu items and sub menu items. Also create a checkable menu item. On clicking a menu Item display a suitable Dialog box.	02*
04	I	Write a program using swing to display a JComboBox in an applet with the items – cricket, football, hockey, tennis	02*
05	I	Write a program to create. – Jtree	02*

Sr. No.	Unit No.	Practical Exercises (Outcomes in Psychomotor Domain)	Approx. Hrs. Required
		– Jtable	
06	II	Write a program making use of Adapter class.	02*
07	III	Write a program to retrieve hostname--using methods in Inet Address class.	02*
08	III	Write a program that demonstrates TCP/IP based communication between client and server.	02*
09	IV	Write an Application program /Applet to make connectivity with database using JDBC, API	02*
10	IV	Write an Application program/Applet to send queries through JDBC bridge & handle result.	02
11	V	Write a servlet for demonstrating the generic servlet class.	02*
12	V	Create a web form which processes servlet and demonstrates use of cookies and sessions	02*
13	V	Develop a simple JSP program for user login form.	02*
14	V	Develop a simple JSP program to display the grade of a student by accepting the marks of five courses.	02
		TOTAL	32

Practicals marked as * are compulsory

7.0 SUGGESTED STUDENT ACTIVITIES:

Other than the Classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course,

1. Discuss different frameworks or tools for advanced java programming. E.g. spring, hibernate, windows builder in eclipse for standalone application etc.
2. Search and Use external jar files in your application.
3. Present your application and discuss various aspects of software e.g. security, efficiency, cost etc.

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Arrange expert seminar of industry person in the area of advanced java programming.
2. More focus should be given on practical work which will be carried out in laboratory sessions. If possible some theory sessions may be conducted in labs so that theory and practice can go hand in hand.

9.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Title of Book	Author	Publication
01	Complete reference	Herbert Schildt	Tata McGraw Hill
02	Unleashed java2 platform	Jamie Jaworski	Techmedia
03	The Complete IDIOT's guide to JAVA2	Michale Morrison	Prentice Hall of India
04	Java Servlets	Karl Moss	Tata McGraw Hill
05	JSP a beginner's guide	Gray Bolinger and Bharti Natarajan	Tata McGraw Hill
06	Craig Walls	"Spring in Action", Manning"	Dreamtech Press, 4th edition

B) Software/Learning Websites:

1. <http://www.oracle.com/technetwork/java/seccodeguide-139067.html>
2. <http://www.tutorialspoint.com/awt/>
3. <http://www.javatpoint.com>
4. <http://docs.oracle.com/javase/5/tutorial/doc/bnafd.html>

C) Major Equipment / Instrument with Broad Specifications:

Sr.No.	Equipment	Specification
01	Desktop Computer	PC Specifications to be followed: Processor: i3 or i5 RAM: 4 GB or better HDD: 1 TB SATA Monitor: TFT LCD OS: Genuine Windows 8 or 10 Professional or Home Premium or Windows 8 or 10 Ultimate Antivirus: User License for three year
02	LCD Projector	Display Type: LCD Light Output: 3200 Lumens
03	Software	JDK 1.8 or above, Any IDE for Java Programming such as Eclipse, JCreator or any other product, Databases like MySql, Oracle, MS-Access, Apache Tomcat Web Server 7 or higher version, Spring Tool Suite, Maven or Gradle

10.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)			
	1	2	3	4	5	6	7	1	2	3	4
CO1	H	H	H	--	--	--	--	L	L	--	--
CO2	--	H	H	M	--	--	--	--	--	--	--
CO3	M	H	H	M	--	--	M	--	--	--	--
CO4	M	H	H	M	--	L	--	--	--	--	--
CO5	--	H	--	H	L	--	M	H	M	H	--

H: High Relationship, M: Medium Relationship, L: Low Relationship.

11.0 SUGGESTED QUESTION PAPER PROFILE:

Unit No	CO	Marks Per Unit	1.35 Times Marks	Question Number Wise Marks						Actual Distribution of Marks
				01	02	03	04	05	06	
I	CO.1	12	16	04	04	04	04	--	04	20
II	CO.2	18	24	08	04	04	04	04	--	24
III	CO.3	18	24	04	04	04	04	04	04	24
IV	CO.4	16	22	04	04	04	--	04	04	20
V	CO.5	16	22	08	--	--	04	04	04	20
	TOTAL	80	108	28	16	16	16	16	16	108

a) Suggested Bitwise Distribution:

Unit No.	I							II							III							IV							V							Total		
CO	1							2							3							4							5									
Marks per Unit	12							18							18							16							16							80		
1.35 Times marks	16							24							24							22							22							108		
Bits	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	Total		
CO	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3	4	4	4	4	4	4	4	5	5	5	5	5	5	5			
Q1	4	-	-	-	-	-	-	4	-	-	-	-	4	-	-	4	-	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	4	4	-	28		
Q2	-	4	-	-	-	-	-	4	-	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	-	-	16			
Q3	4	-	-	-	-	-	-	4	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	-	-	16			
Q4	-	-	4	-	-	-	-	-	-	4	-	-	-	-	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4	-	-	-	-	16			
Q5	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	-	4	-	-	-	4	-	-	-	-	-	16			
Q6	4	-	-	-	-	-	-	-	-	-	-	-	-	-	4	-	-	-	-	-	-	-	-	-	4	-	-	-	-	-	4	-	-	-	16			
Sub Total	20							24							24							20							20							108		
TOTAL																																						108

